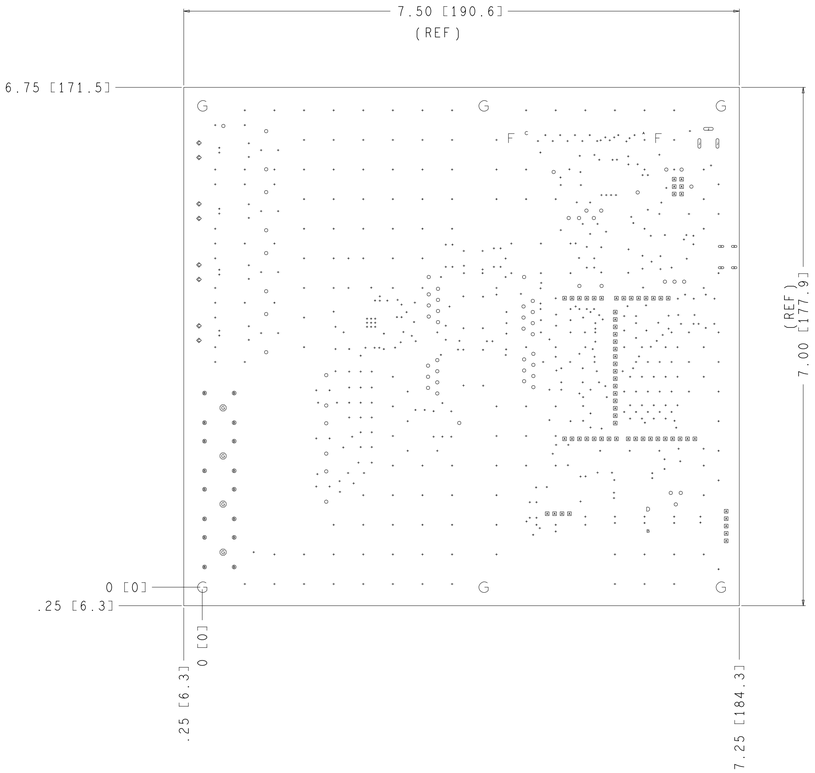
REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE	18OCT21	J.MENDIA
A1	AS PER ECR-117961	13DEC23	S.TRIMBLE
B	AS PER ECR-118624	20DEC23	S.TRIMBLE
C	AS PER ECR-121732	23JUL24	A.MUKHERJEE

HOLE TOLERANCE

UNLESS SPECIFIED
PLATED: +/- 3 MILS
NON PLATED: +/- 2 MILS

DRILL CHART: TOP to BOTTOM				
FINISHED HOLES IN MILS				
FIGURE	SIZE	PLATED	QTY	TOLERANCE/NOTES
·	20.0	PLATED	554	DIA MAX SEE NOTE 15
○	40.0	PLATED	64	
⊠	45.0	PLATED	63	
⊛	50.0	PLATED	16	
◈	60.0	PLATED	8	
◎	87.0	PLATED	4	
⋆	28.0	NON-PLATED	1	
⊛	39.0	NON-PLATED	1	
c	43.0	NON-PLATED	1	
D	51.0	NON-PLATED	1	
F	125.0	NON-PLATED	2	
G	144.0	NON-PLATED	6	
∞	75.0x28.0	PLATED	4	
⧘	130.0x39.0	PLATED	2	
⧙	130.0x39.0	PLATED	1	



PRIMARY SIDE

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES DECIMALS FRACTIONS ANGLES .XX +-.010 +-1/32 +- 2 .XXX +-.005 .XXXX +-.0050	APPROVAL	DATE
	TEMPLATE ENGINEER	
	HARDWARE SERVICES E.CONDA	18OCT21
	HARDWARE SYSTEMS	
MATERIAL	TEST ENGINEER	
	COMPONENT ENGINEER M.YAN	18OCT21
	TEST PROCESS N/A	N/A
	HARDWARE RELEASE E.CONDA	18OCT21
FINISH	DESIGNER P.CORDERO	18OCT21
	PTD ENGINEER J.MENDIA	18OCT21
	CHECKER	
DO NOT SCALE DWG		

ANALOG DEVICES
WWM DIVISION
804 WOBURN STREET
WILMINGTON, MA 01887

TITLE
FABRICATION
ADE9430
CUSTOMER EVALUATION Z

SIZE	FSCM NO	DRAWING NUMBER	REV
D	24355	09-068967	C

SCALE 1/1 SHEET 1 OF 2

ADE9430 CHARACTERISATION BOARD

REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE	18OCT21	J.MENDIA
A1	AS PER ECR-117961	13DEC23	S.TRIMBLE
B	AS PER ECR-118624	20DEC23	S.TRIMBLE
C	AS PER ECR-121732	23JUL24	A.MUKHERJEE

ASSEMBLY NOTES:

1. BOARD ACCEPTABILITY PER ANALOG DEVICES, INC. SPECIFICATION TST00119 (LATEST REVISION).

2. REPAIRS PER IPC-7711/21(LATEST REVISION) ARE ALLOWED.

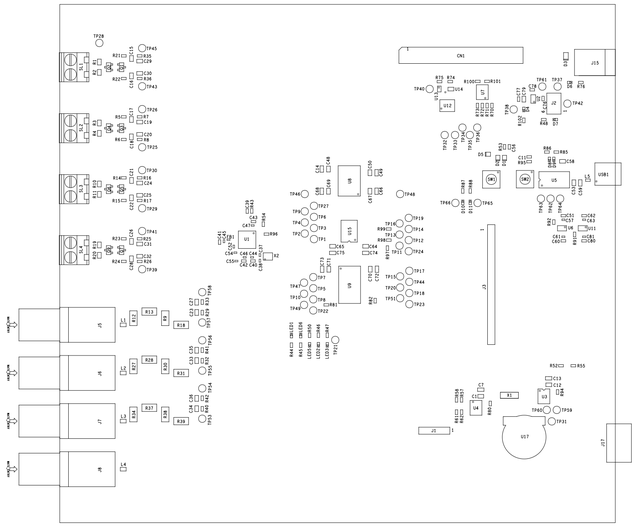
3. REPAIRS ARE NOT ALLOWED IN SOLDERMASK FREE AREAS ON EITHER SIDE OF THE BOARD.

4. RoHS COMPLIANCE: ASSEMBLY VENDOR SHOULD ASSURE COMPLIANCE WITH LEAD-FREE AND RoHS PCB ASSEMBLY STANDARDS (EU RoHS DIRECTIVE 2002/95/EC).

5. SPECIFIC COMPONENTS SHOULD BE INSTALLED WITH THEIR CORRESPONDING MECHANICAL PARTS LISTED ON THE MECHANICAL PARTS SECTION OF THE BOM.

Qty	WW_PN	DESCRIPTION	VALUE	JEDEC_TYPE
1	E000939	CONN-PCB 2.1MM DC POWER M016045 QTY 1	DC-10A (FC68148)	CNCECDC-10

6. INSERT SHUNT (M029337) ON PIN 1-6 (5V_MCU) OF J2 CONNECTOR.

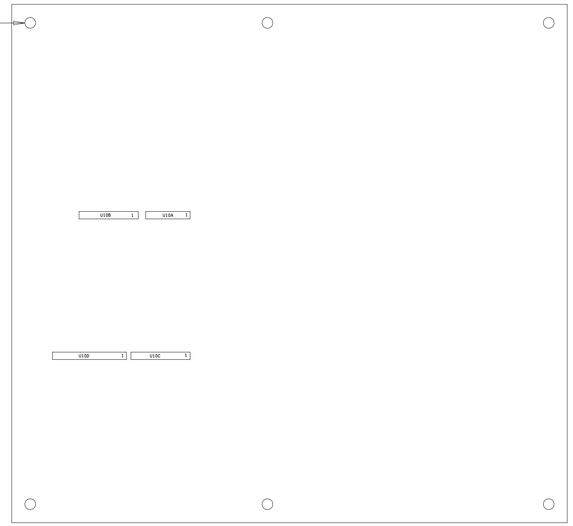


PRIMARY SIDE

ADE9430 CHARACTERISATION BOARD

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES DECIMALS FRACTIONS ANGLES .XX +-.010 +-1/32 +- 2 .XXX +-.005 .XXXX +-.0050	APPROVAL	DATE	ANALOG DEVICES WWM DIVISION 804 WOBURN STREET WILMINGTON, MA 01887
	TEMPLATE ENGINEER		
	HARDWARE SERVICES E.CONDA	18OCT21	
	HARDWARE SYSTEMS		TITLE ASSEMBLY ADE9430 CUSTOMER EVALUATION Z
MATERIAL	TEST ENGINEER		
	COMPONENT ENGINEER M.YAN	18OCT21	
	TEST PROCESS N/A	N/A	
	HARDWARE RELEASE E.CONDA	18OCT21	
FINISH	DESIGNER P.CORDERO	18OCT21	SIZE D / FSCM NO 24355 / DRAWING NUMBER 01-068967 / REV C
	PTD ENGINEER J.MENDIA	18OCT21	
	CHECKER		
DO NOT SCALE DWG			SCALE 1/1 / SHEET 1 OF 2

INSTALL STANDOFF,SCREW AND WASHERS, 6 PLACES



ADE9430 CHARACTERISATION BOARD

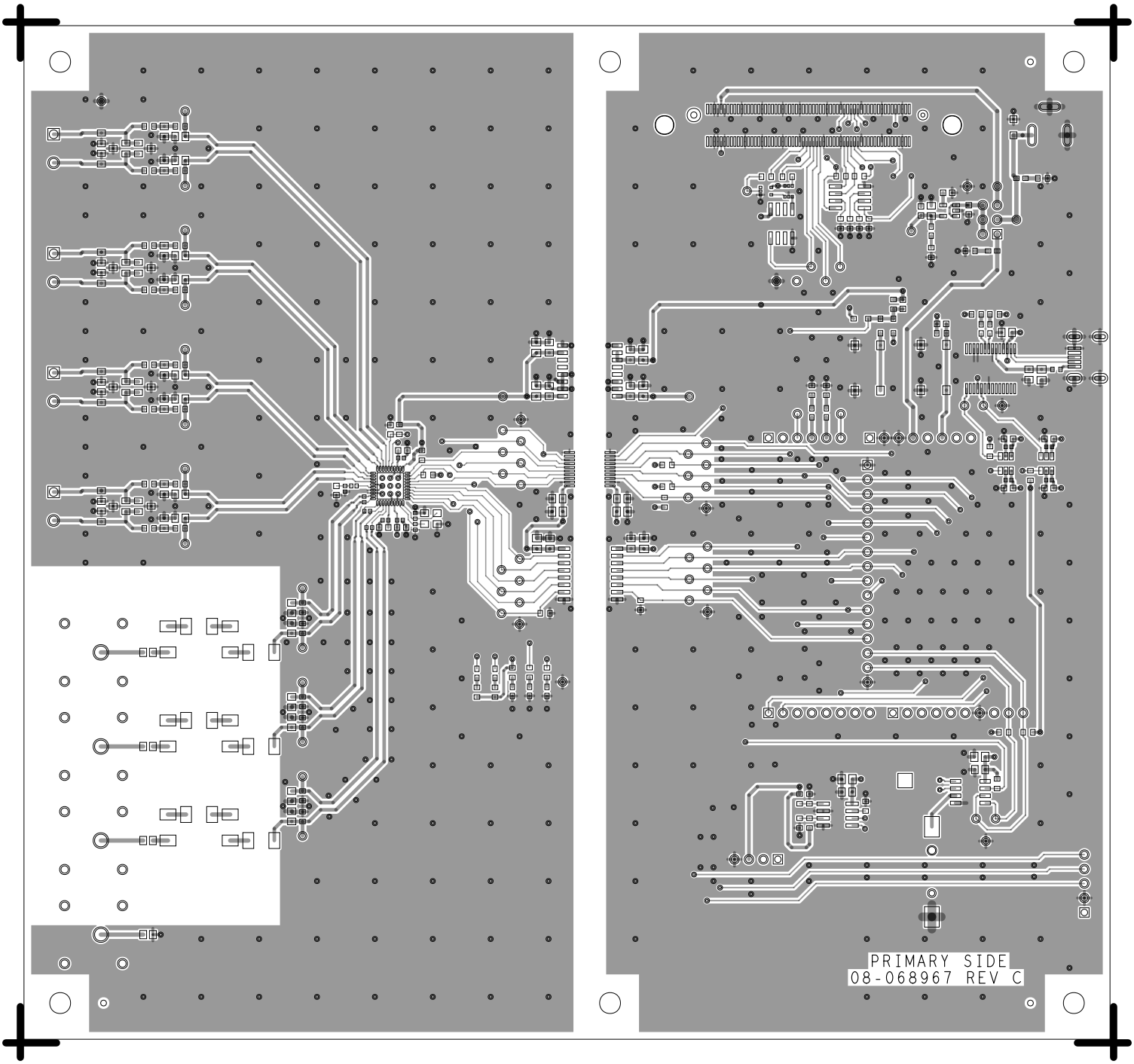
SECONDARY SIDE

ANALOG DEVICES

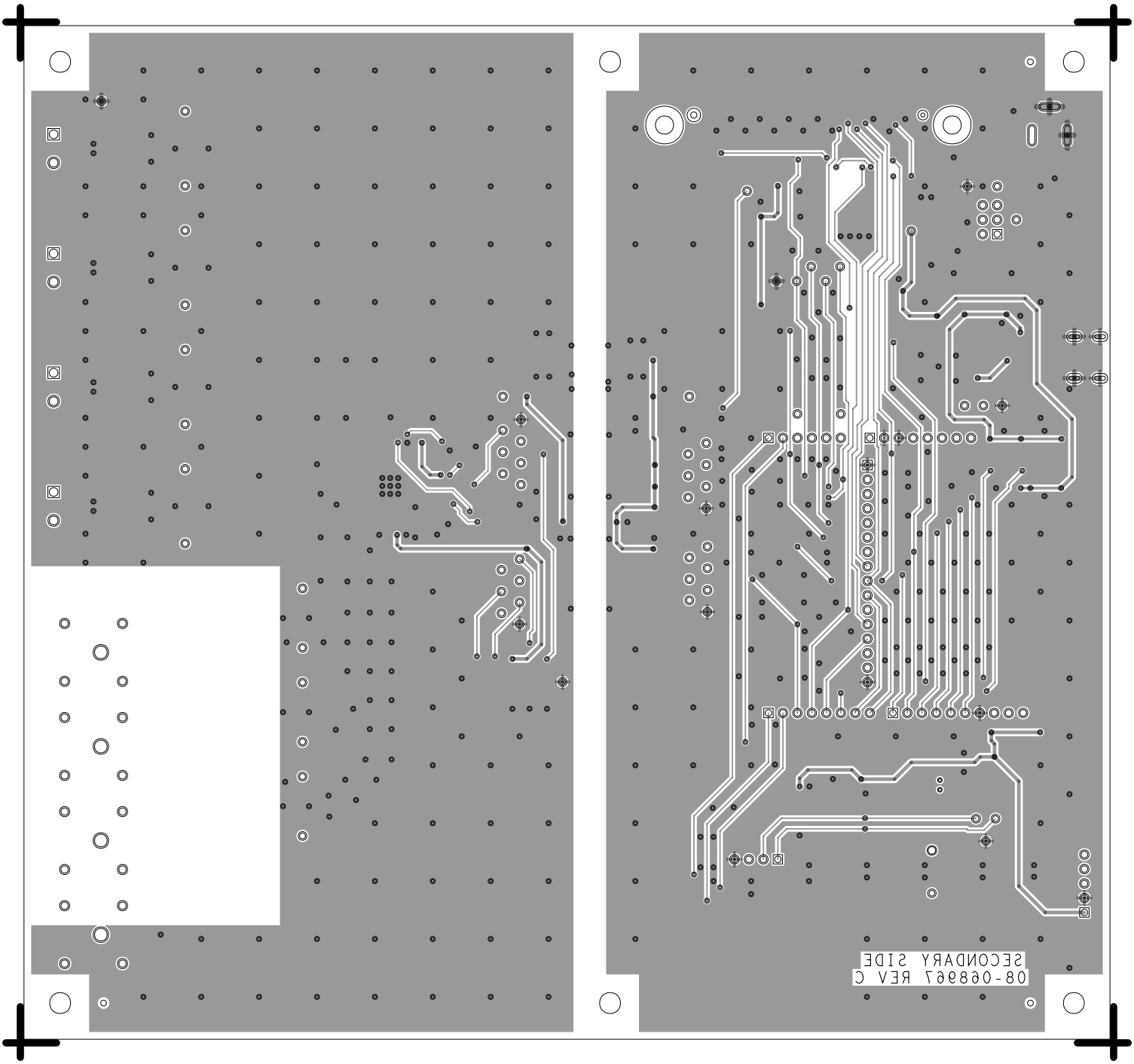
WWM DIVISION
804 WOBURN STREET
WILMINGTON, MA 01887

SIZE	FSCM NO	DRAWING NUMBER	REV
D	24355	01-068967	C
SCALE 1/1			SHEET 2 OF 2

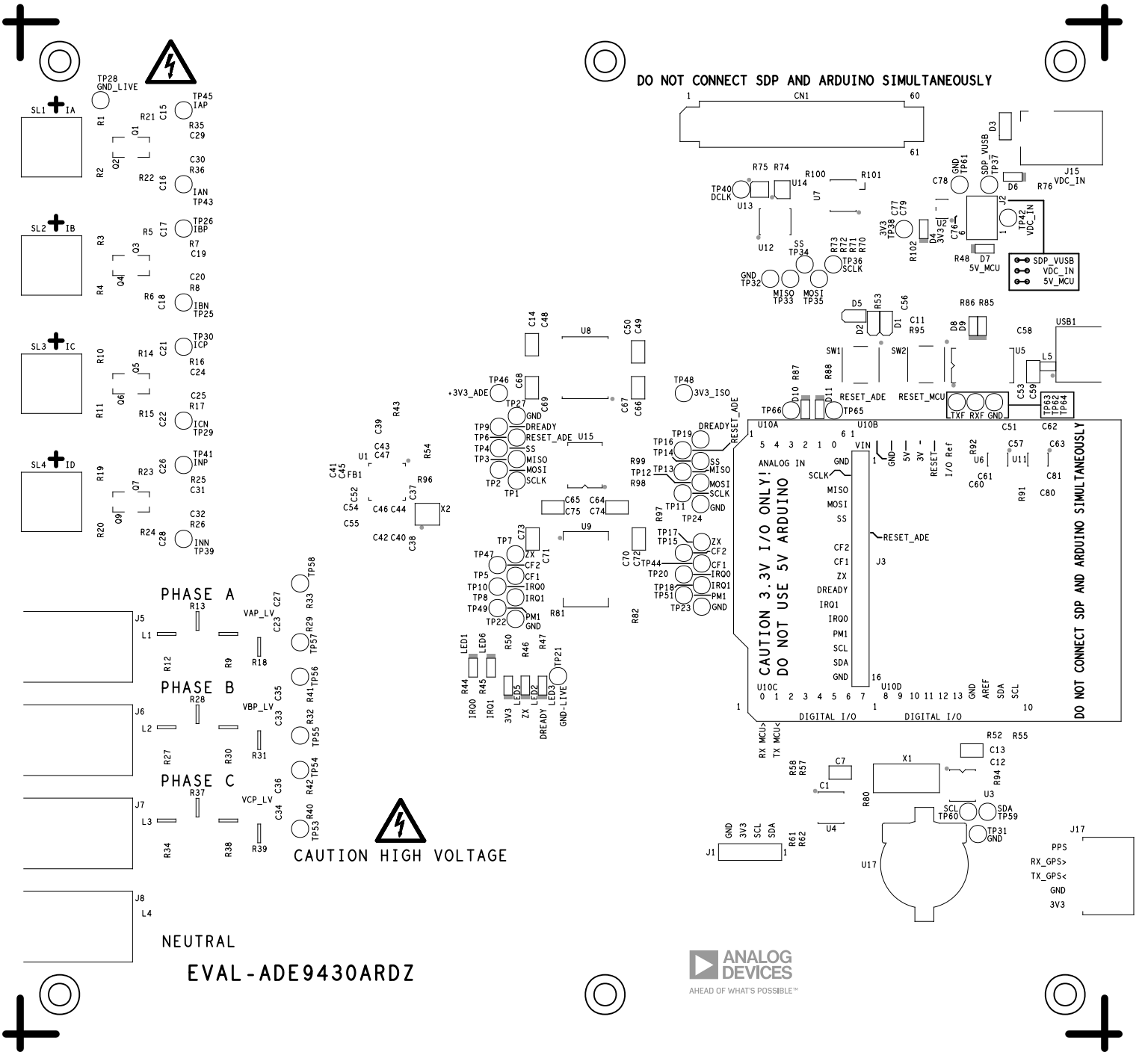
L1 PRIMARY
08-068967-01
REV C



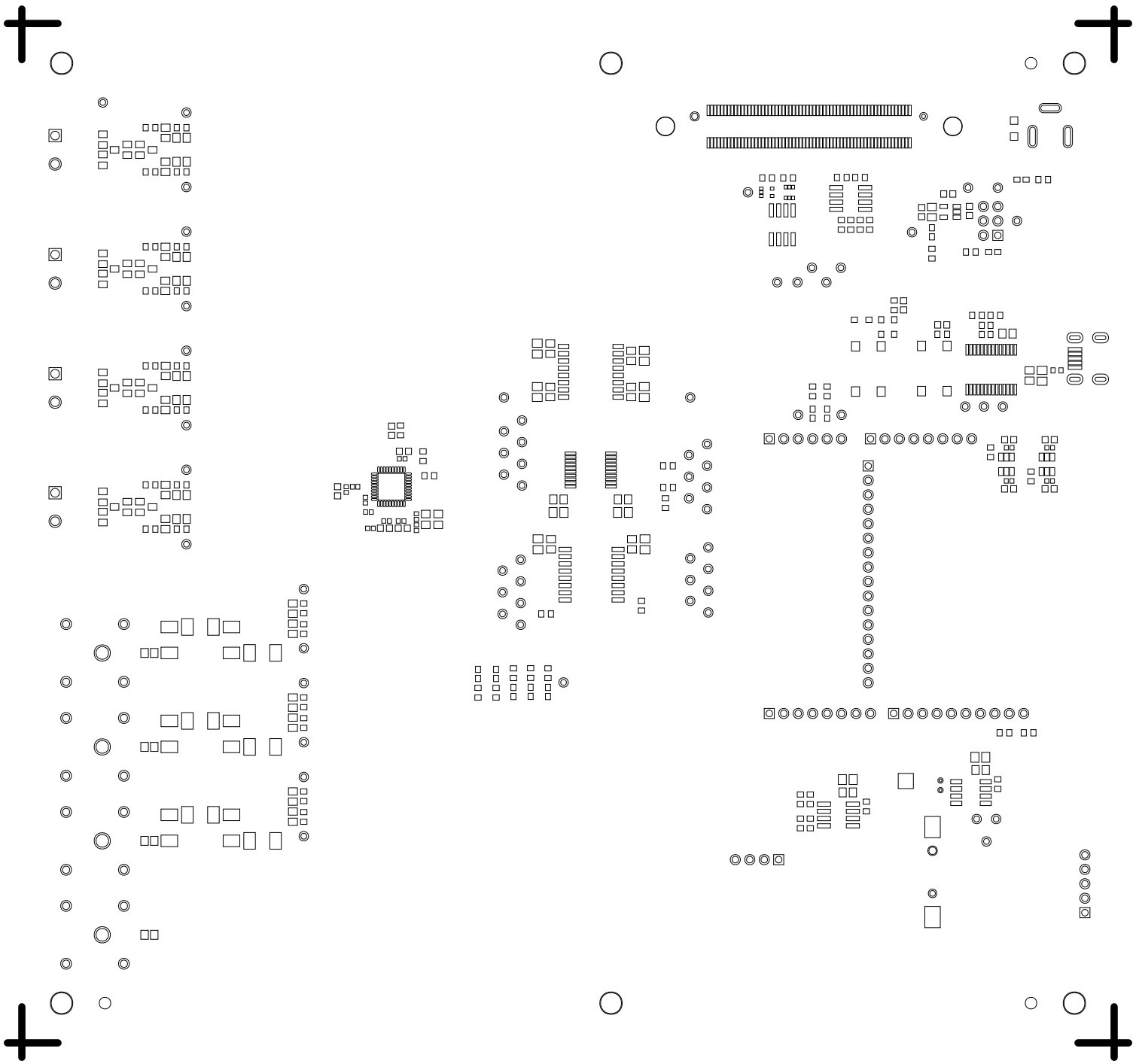
L2 SECONDARY
08-068967-02
REV C



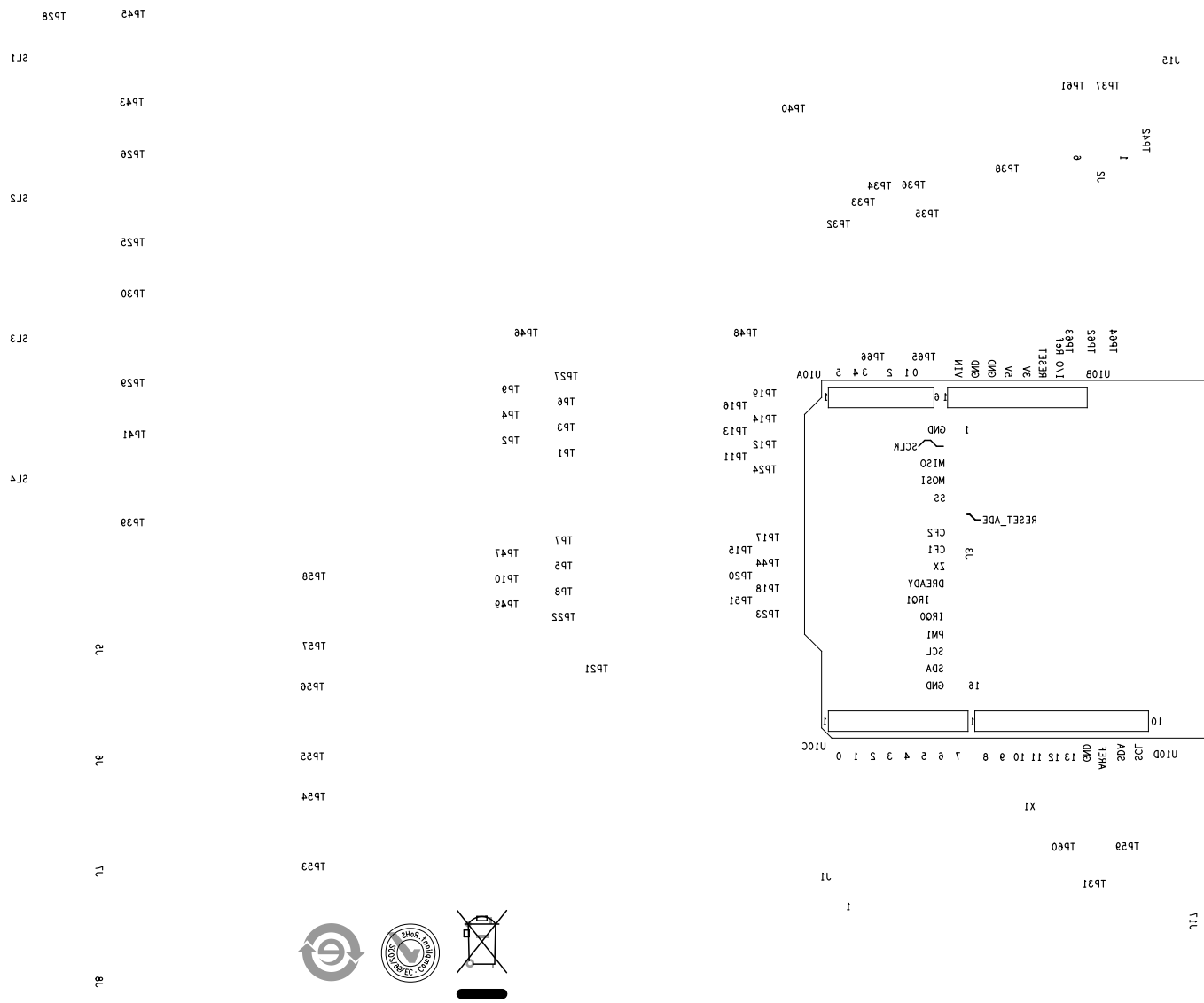
SILKSCREEN PRIMARY
08-068967-03
REV C



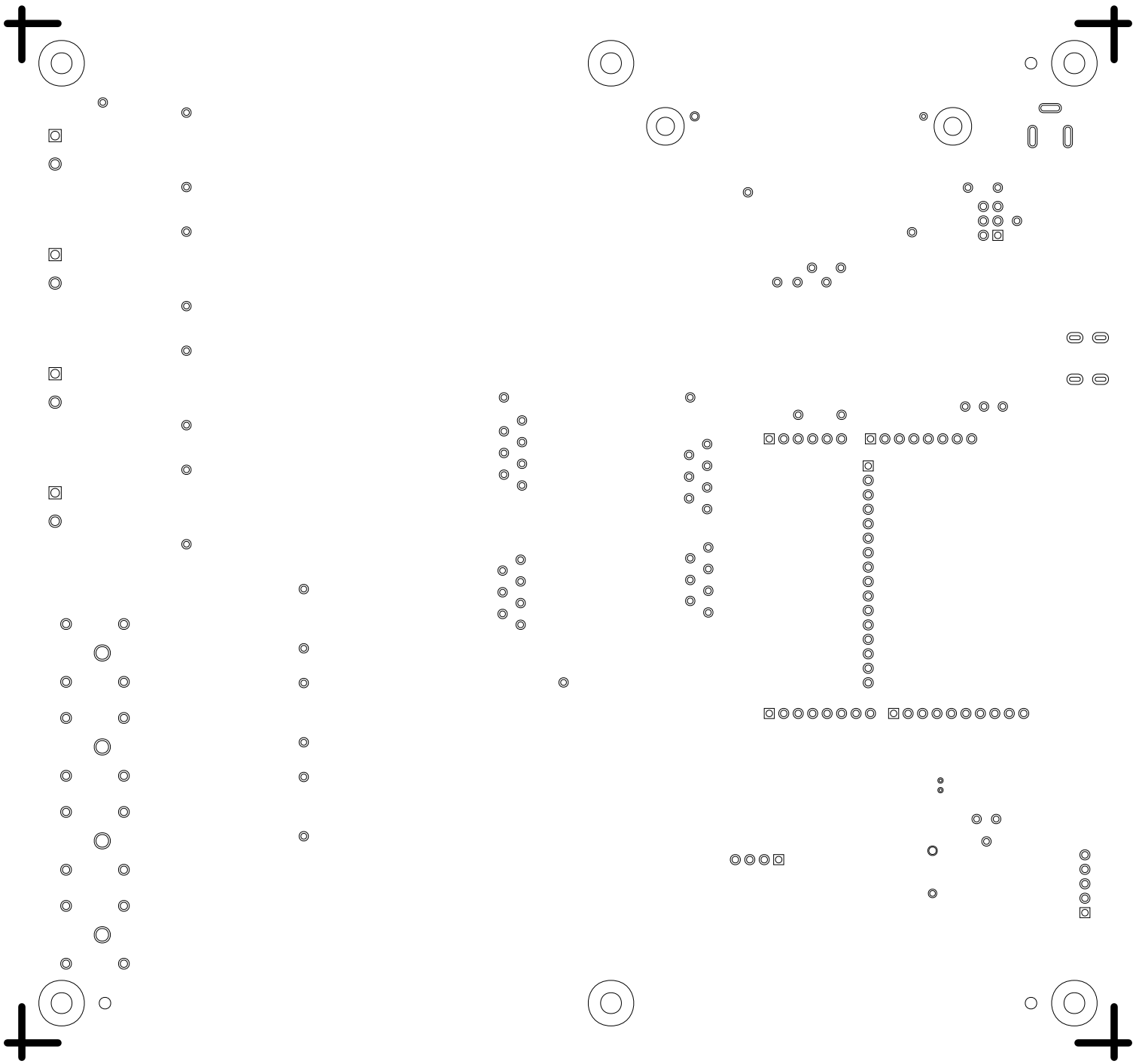
SOLDERMASK PRIMARY
08-068967-04
REV C



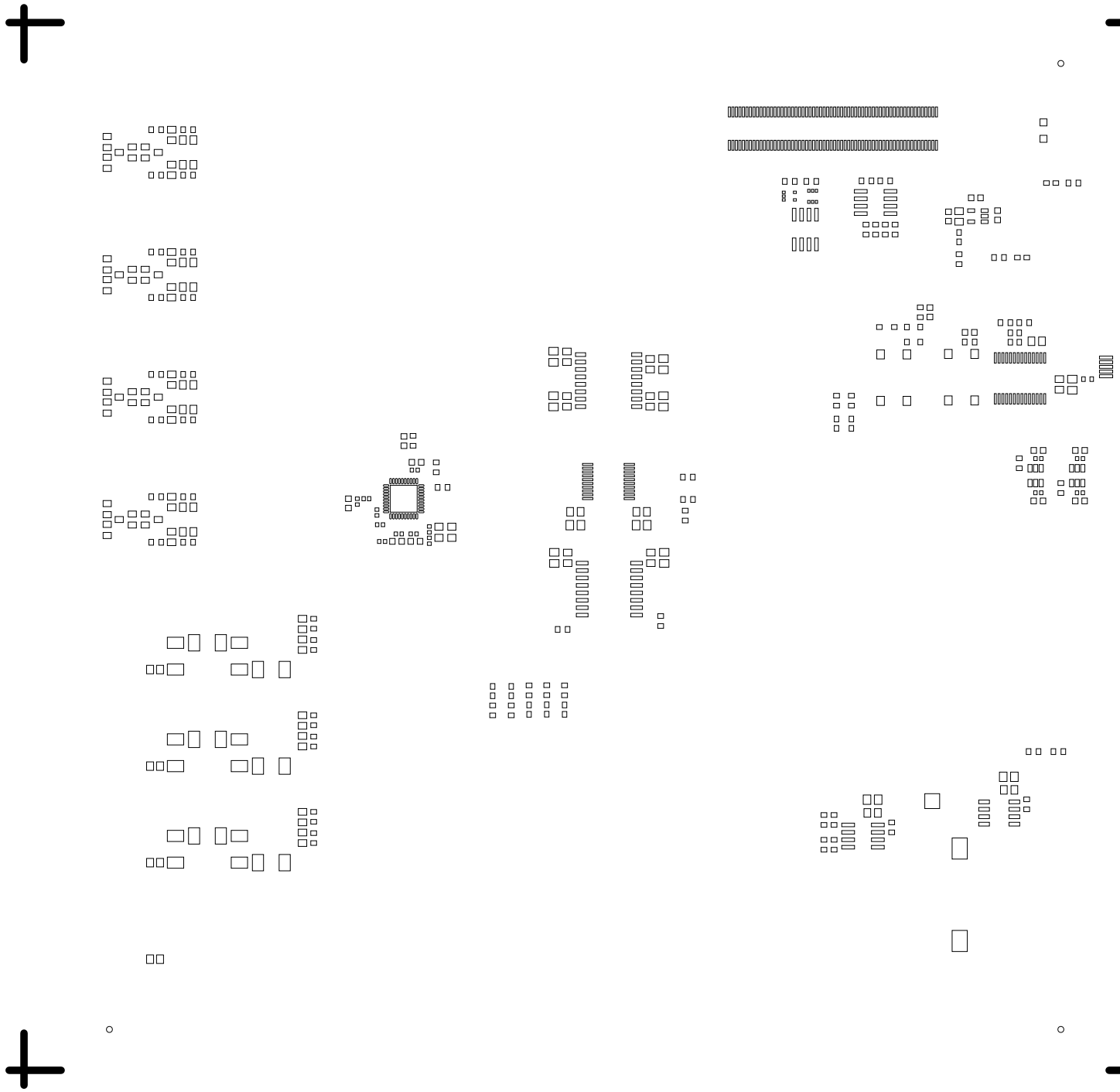
SILKSCREEN SECONDARY
08-068967-05
REV C



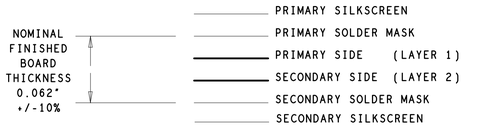
SOLDERMASK SECONDARY
08-068967-06
REV C



PASTEMASK PRIMARY
08-068967-07
REV C



2 LAYER STACKUP



SPECIFICATIONS:

ROHS COMPLIANCE NOTE: HOMOGENOUS MATERIALS IN THIS BOARD SHALL BE COMPLIANT THE EU RoHS DIRECTIVE 2002/95/EC

MATERIALS; ALL LAMINATES AND BONDING MATERIALS SHOULD BE SELECTED FROM IPC-4101 OR IPC-4103, MINIMUM Tg>170degC, Td>300degC, U.L. RATING OF 94 V-0

MATERIAL FAMILY; ISOLA 370HR

CLADDING; EXTERNAL LAYERS .5 OZ. COPPER, OVERPLATE TO 1.5 OZ.

NOTE: IF THE LAYER STACKUP CONFLICTS WITH THE ABOVE CLADDING SPECIFICATIONS THEN THE LAYER STACKUP SHALL TAKE PRECEDENCE.

SOLDER MASK; SHALL BE LIQUID PHOTOIMAGEABLE (LPI) APPLIED ON BOTH SIDES OVER BARE COPPER OR GOLD AND SHALL MEET IPC-SM-840 (LATEST REV.) CLASS 3. COLOR GREEN.

SILK SCREEN; SHALL BE PERMANENT NON-CONDUCTIVE EPOXY INK, COLOR: WHITE SYNTHETIC INKJET PRINTING ALLOWED FOR DENSE BOARDS, COLOR: WHITE

SURFACE FINISH; ENIG (Electroless Nickel/Immersion Gold) PER IPC-4552 LATEST REVISION

INTENTIONAL SHORTS; IF SUPPLIED DATA INCLUDES A FILE "READ_ME.2", THEN INTENTIONAL NET SHORTS EXIST. CUSTOMER REVIEW AND APPROVAL IS REQUIRED IF SUPPLIED DATA REPORTS ANY CONDITION THAT DOES NOT MATCH "READ_ME.2" FILE PROVIDED.

TEST REQUIREMENTS; 100% NETLIST ELECTRICAL VERIFICATION USING CUSTOMER SUPPLIED IPC-D-356 NETLIST FOR OPENS AND SHORTS WHEN "GERBER DATA" IS PROVIDED. THIS VERIFICATION ALSO REQUIRED FOR "ODB++" DATA PER EMBEDDED NETLIST.

REQUIREMENTS:

1. REFER TO IPC-6010 SERIES (LATEST REV.), CLASS 2 FOR FABRICATION UNLESS OTHERWISE SPECIFIED.
2. ACCEPTABILITY PER ANALOG DEVICES, INC. SPECIFICATION TST00115, (LATEST REVISION.)
3. MODIFICATIONS TO THE ARTWORK ARE NOT ALLOWED WITHOUT WRITTEN AUTHORIZATION.
4. HOLE PATTERN TOLERANCES FOR UNDIMENSIONED HOLES SHALL BE A DIAMETER OF 0.005 INCHES FROM THEIR TRUE POSITION.
5. PLATED HOLE WALL THICKNESS SHALL NOT BE LESS THAN 0.001 INCH MINIMUM AVERAGE, WITH NO READING LESS THAN .0008 BY CROSS SECTION.
6. HOLE DIAMETERS APPLY AFTER PLATING.
7. FINISHED CONDUCTOR WIDTHS SHALL NOT BE REDUCED FROM THE NOMINAL INDICATED ON THE MASTER PATTERN, BY MORE THAN THE CONDUCTOR THICKNESS.
8. MINIMUM DESIGN LINE WIDTH IS .004 INCH.
9. MINIMUM DESIGN SPACING IS .005 INCH.
10. NON-FUNCTIONAL PAD REMOVAL FROM INNER SIGNAL LAYERS MAY BE PERFORMED AFTER CUSTOMER APPROVAL.
11. IF PAD SIZES PROVIDED ARE NOT LARGE ENOUGH TO MAINTAIN ANNULAR RING REQUIREMENT, MFGR. MAY REQUEST APPROVAL TO TEAR DROP PADS TO MAINTAIN ANNULAR RING. (AT PAD TO TRACE INTERSECTION ONLY AND ELECTRICAL INTEGRITY MUST BE MAINTAINED.)
12. THIEVING MAY BE ADDED TO COMPENSATE FOR LOW COPPER DENSITY AREAS ON THIS DESIGN ONLY AFTER REVIEW AND APPROVAL FROM THE CUSTOMER:
    A. THIEVING TO CARD EDGE, FIDUCIALS, NON-PLATED THROUGH HOLES, ALL OTHER FEATURES TO BE 0.200 INCH MINIMUM.
    B. THERE SHALL BE NO THIEVING IN ANY AREAS FREE OF SOLDER MASK OR INTERNAL COPPER PLANES.
13. MFGR. TO LEGIBLY ETCH OR STAMP/SCREEN WITH PERMANENT NON-CONDUCTIVE INK ON SECONDARY SIDE IN A CLEAR AREA UNLESS OTHERWISE INDICATED;
    A. U.L. CODE-FLAMMABILITY RATING
    B. DATE CODE (STAMP).
    C. LOT NUMBER
    D. MFGR LOGO
    E. SUCCESSFUL ELECTRICAL TEST.
14. REPAIRS PER IPC-7711/21 (LATEST REV.) ARE ALLOWED. REPAIRS ARE NOT ALLOWED IN ANY AREA DEFINED ON GOLD_PRM AND/OR GOLD_SEC ARTWORK LAYERS WHEN PROVIDED IN GERBER OR ODB++ DATA.
15. THRU VIAS FILLED WITH NON-CONDUCTIVE EPOXY AND PLATED OVER.

PRIMARY SIDE

ANALOG DEVICES
WWM DIVISION
804 WOBURN STREET
WILMINGTON, MA 01887

SIZE	FSCM NO	DRAWING NUMBER	REV
D	24355	09-068967	C
SCALE 1/1			SHEET 2 OF 2

ADE9430 CHARACTERISATION BOARD